

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	English-only emotion detection	Excludes non-English speaking	High
2	No user authentication or	Cannot track individual student	High
3	Gemini free tier limit (1500	App unusable after daily quota	High
4	BiLSTM + BERT weights not	Full ML pipeline unavailable on	Medium
5	Single user text input only	Cannot analyze audio, images,	Medium
6	Static keyword dictionary	New slang or domain-specific	Low
7	CSV-based logging	Not scalable beyond ~100K interactions	Low

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
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S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	10 concurrent Streamlit Cloud users	Migrate to FastAPI backend + React frontend on cloud VMs (AWS/GCP)
2	Data Storage	Flat CSV file	Migrate to PostgreSQL or Firebase Realtime DB for structured, indexed storage
3	Performance	Gemini API latency ~2-3s	Cache Gemini responses for identical inputs using Redis
4	Model Deployment	Gemini-only on cloud	Deploy BiLSTM + BERT as REST microservices using FastAPI + Docker on a VPS
5	Security	Gemini key in Streamlit Secrets	Use OAuth2 for user auth; store keys in AWS Secrets Manager

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
6	Multilingual Support	English only	Integrate Google Translate API for preprocessing; retrain on multilingual dataset
7	Analytics	Static Plotly charts	Build a real-time dashboard with WebSocket streaming (Plotly Dash / Grafana)

Future Feature Roadmap

Priority	Feature	Description	ETA
High	User Login & Profiles	Track individual student	Q3 2026
High	Multilingual Support	Detect emotions from Hindi,	Q3 2026
Medium	Voice Input	Analyze emotion from audio recordings via speech-to-text +	

NLP

			Q4 2026
Medium	Teacher Dashboard	Aggregated class-level emotion	

Q4 2026

Priority	Feature	Description	ETA
Medium	Mobile App	React Native app wrapping the	Q4 2026
Low	Facial Expression Fusion	Multimodal detection combining	2027
Low	LMS Integration	Plugin for Moodle / Google	2027
Low	Custom Model Fine-Tuning	Let institutions train on their own	2027
		student data	

Scalability Architecture (Future State)

Student (Web / Mobile)

Load Balancer (Nginx)

FastAPI Backend
(Docker Containers)

BiLSTM Service
BERT Service
Gemini Client

PostgreSQL DB
Redis Cache
AWS S3 (Models)